Business Statistics

Module 7
Video 7.1
Introduction: Measures of Central Tendency

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Agenda

- Measures of Central Tendency- Introduction
- Mean, Median, and Mode- Grouped and Ungrouped Data
- Central Tendency of Categorical Data

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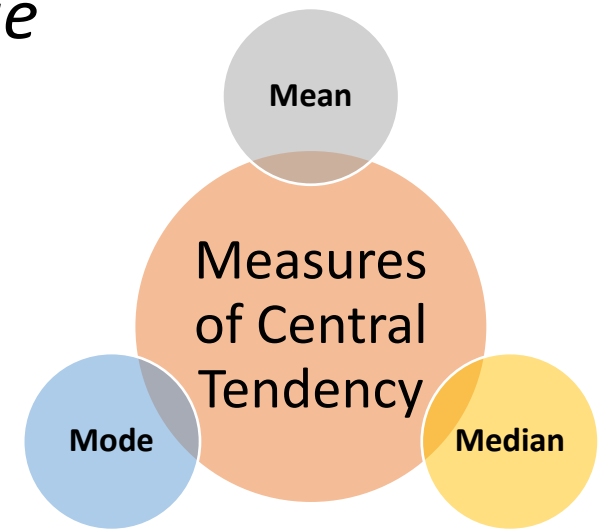
Measures of Central Tendency

Measure of Central Tendency

A single value that describes the central position of the data points- it indicates where most values in the data lie

This provides a good summary of the data as a ***single numeric point*** and can be used to describe the data rather than presenting the whole data itself



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Some Interesting Averages

- India (Men): 5ft 4.9 inches
- Latvia (Men):5ft 9 inches

Average Height

- Tesla: $3,029
- Porsche: $17,666

Average Profit Per Car

- Rohit Sharma: 58.11
- Sachin Tendulkar: 48.29

Average Run Rate

Can we get a picture of the data just by looking at these numbers?
What conclusions can we draw?

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Mean, Median, and Mode

Arithmetic Mean

$$\bar{x} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Median

The middle value when the data points are arranged in order

Mode

The most frequently occurring value

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Mean, Median, Mode

When might you need to use measures of central tendency?

- Graduates of which universities earn the highest salaries?

Average/Median Salary of Graduates

- Which is the most popular social media channel?

Mode of the social media preferences

- Which company shares should an investor buy?

Average returns on the shares over the year

- Which equipment should a factory owner buy based on quality ratings and prices?

Median- price not too high quality not too low

And there could be many other cases where you need to summarise the data with a single number!

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Central Tendency: Ungrouped Data

The data is given as individual values (ungrouped):

$$Mean\ (\bar{x}) = \frac{\sum x_i}{n}$$

$$Median = \frac{n+1}{2} th\ data\ point, when\ the\ data\ is\ arranged\ in\ order$$

$$Mode = The\ value\ which\ is\ repeated\ most\ frequently$$

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Example 1

Example: The ages of a employees of a small company are recorded

S. No	Age
1	28
2	45
3	91
4	68
5	47
6	28
7	37
8	78

*What is the **mean** age?*

*What is the **median** age?*

*What is the **modal** age?*

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Example 1: Continued

S. No	Age
1	28
2	28
3	37
4	45
5	47
6	68
7	78
8	91

$$Mean = \frac{28 + 28 + \cdots + 91}{8} = 52.75 \text{ years}$$

$$Median = \frac{n+1}{2} th \text{ value i. e. } 4.5\text{th value} = \frac{45+47}{2} = 46\ years$$

$$Mode = 28\ years$$

The central measures, Mean and Median give us the indication that the group is middle aged

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Central Tendency: Ungrouped Data with Frequency

The data values are given along with their frequency

$$Mean\ (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = \frac{\sum f_i x_i}{N}, where\ f_i\ is\ the\ frequency\ of\ x_i$$

$$Median = \frac{n+1}{2} th\ data\ point, when\ the\ data\ is\ arranged\ in\ order$$

$$Mode = The\ value\ with\ the\ highest\ frequency$$

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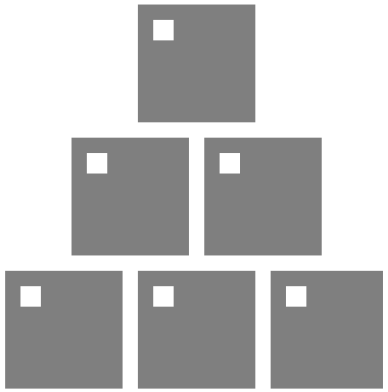
Example 2

The total units of a product that the customers order from a manufacturer in the year 2019 are tabulated below.

Units Sold	Frequency
12	3
13	6
15	1
17	3
18	8
23	10
26	2
27	4



*What is the **mean** number of orders?*

*What is the **median** number?*

*What is the **modal** number of orders?*

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Example 2

Units Sold	Frequency
12	3
13	6
15	1
17	3
18	8
23	10
26	2
27	4

19th Value

$$Mean = \frac{(12 \times 3) + (13 \times 6) + \cdots + (27 \times 4)}{3 + 6 + \cdots + 4} = \frac{714}{37} = 19.3$$

$$Median = \frac{N+1}{2} th \text{ value i.e. } 19\text{th value} = 18$$

$$Mode = 23$$

The average number of orders that the manufacturer receives is 19.3

Exactly 50% of the orders received were <18

The most frequent number of orders received is 23

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Central Tendency: Grouped Data

The data values are given in grouped form with their classes and frequency

$$Mean\ (\bar{x}) = \frac{\sum f_i m_i}{\sum f_i} = \frac{\sum f_i m_i}{N}$$

$$f_i\ is\ the\ frequency\ of\ class\ i$$

$$m_i is\ the\ midpoint\ of\ class\ i, m_i = \frac{Lower\ Limit_i + Upper\ Limit_i}{2}$$

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Central Tendency: Grouped Data

$$Median = l + \frac{\left(\frac{N}{2} - c.f\right)}{f} \times c$$

l is the lower limit of the median class

c is the width of the median class

f is the frequency of the median class

$c.f$ is the cumulative frequency of the preceding class

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Central Tendency: Grouped Data

$$Mode = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c.i$$

$$L \text{ is the lower limit of the modal class}$$
$$f_0 \text{ is the frequency of the pre} - \text{modal class}$$
$$f_1 \text{ is the frequency of the modal class}$$
$$f_2 \text{ is the frequency of the succeeding modal class}$$
$$c.i \text{ is the class interval}$$

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Example 3

The revenue generated by different divisions of a company (in thousands of dollars) is tabulated below.

Revenue	Frequency
0-5	3
5-10	6
10-15	1
15-20	3
20-25	4
25-30	7
30-35	2
35-40	4

*What is the **mean** revenue?*

*What is the **median** revenue?*

*What is the **modal** revenue?*

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Example 3: Continued

Revenue	Frequency	Cumulative Frequency	Mid-Point
0-5	3	3	2.5
5-10	6	9	7.5
10-15	1	10	12.5
15-20	3	13	17.5
20-25	4	17	22.5
25-30	7	24	27.5
30-35	2	26	32.5
35-40	4	30	37.5

$$Mean$$

$$= \frac{(3 \times 2.5) + (6 \times 7.5) + \cdots + (4 \times 37.5)}{3 + 6 + \cdots + 4}$$

$$= \frac{615}{30} = 20.5$$

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Example 3: Continued

Revenue	Frequency	Cumulative Frequency	Mid-Point
0-5	3	3	2.5
5-10	6	9	7.5
10-15	1	10	12.5
15-20	3	13	17.5
20-25	4	17	22.5
25-30	7	24	27.5
30-35	2	26	32.5
35-40	4	30	37.5

$$Median$$

Median class is the class that contains the $\frac{N}{2}$ *th data point i.e. the 15th data point*

$$Median = l + \frac{\left(\frac{N}{2} - c.f\right)}{f} \times c$$

$$= 20 + \frac{15 - 13}{17} \times 5 = 20.59$$

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Example 3: Continued

Revenue	Frequency	Cumulative Frequency	Mid-Point
0-5	3	3	2.5
5-10	6	9	7.5
10-15	1	10	12.5
15-20	3	13	17.5
20-25	4	17	22.5
25-30	7	24	27.5
30-35	2	26	32.5
35-40	4	30	37.5

$$Mode$$

Modal class is the one with the highest frequency

$$Mode = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c.i$$

$$= 25 + \frac{7 - 4}{2 \times 7 - 4 - 2} \times 5 = 26.88$$

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Example 3: Continued

The average revenue generated by the divisions is 20.5 thousand dollars

50% of the divisions generated a revenue > 20.59 thousand dollars

The most frequently generated revenue is 26.88 thousand dollars

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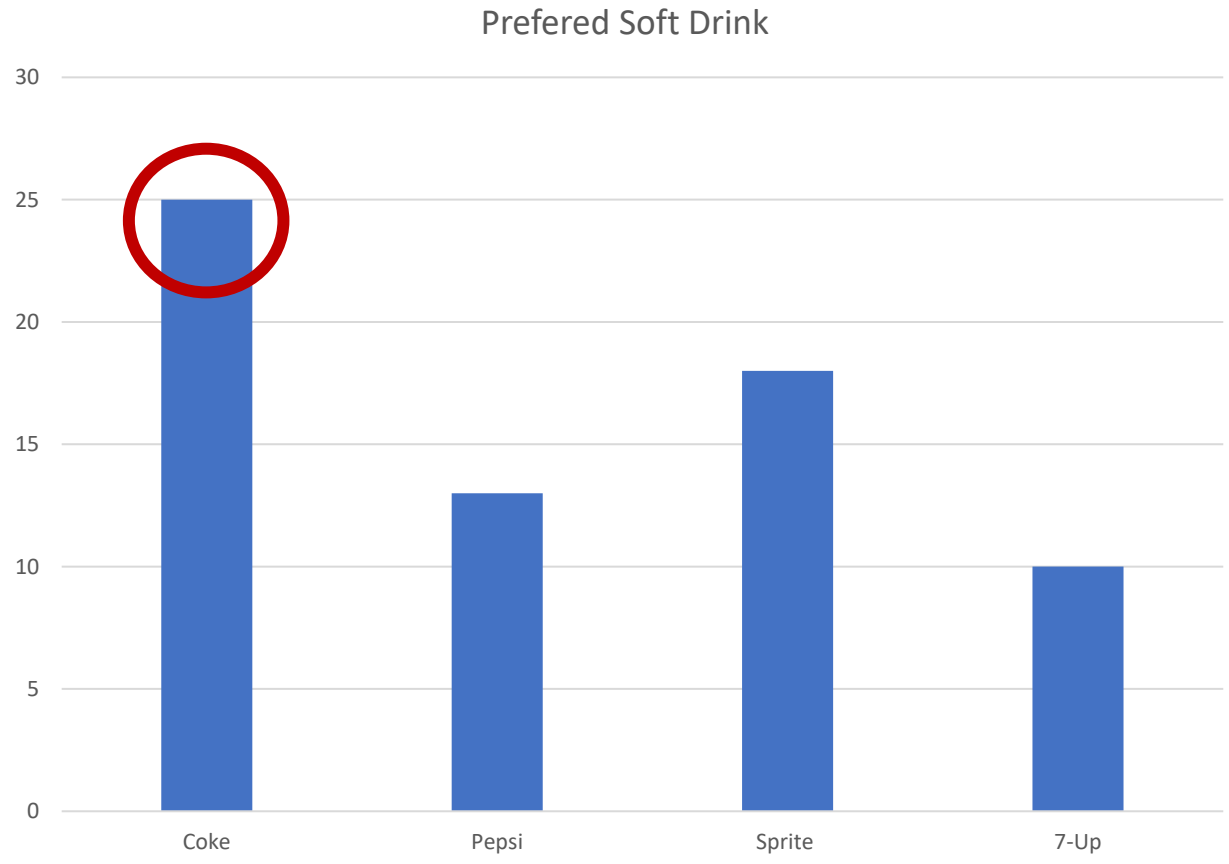
Central Tendency of Categorical Data

Is there some measure of central tendency in categorical data?

- The Bar chart represents categorical data
- The data does not have a **“central value”**
- We can use mode- the most common value in the data



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Summary

- Measures of Central Tendency: Single measure for the central location of the data
  - *Mean*
  - *Median*
  - *Mode*
- Mean: $\bar{x} = \dfrac{\text{Sum of observations}}{\text{Number of observations}}$
- Median: The middle value in the data
- Mode: The most commonly occurring value in the data

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